Lecture 28 — Composition and matrix powers

Wed Oct 28

Last updated: October 8, 2026.

Boyd video (optional): Lecture 29 — matrix powers & QR · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 10.2–10.3. Related objectives: Lectures. Schedule.

Note: We return to the Week 9 dynamics lab from a new viewpoint — matrix powers package repeated steps into one matrix.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Combine multistage processes using a matrix product.
2. Connect $A^k x$ to repeated dynamics.
3. Interpret selected entries of a matrix product or power.
4. Recognize how network and transition information accumulates through powers.

Warm-up

A two-stage process on $\mathbb{R}^2$: first apply

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

(shear), then apply

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(90° CCW). Write the single matrix M for "B then A" as a product, and compute $M(1, 0)$.

Fill in

Composition. If $g(x) = Bx$ and $f(y) = Ay$ (compatible sizes), then

$$(f \circ g)(x) = f(g(x)) = _____.$$

The matrix of the composed map is the _____ AB (apply B first). Multistage processes: write each stage as a matrix, then multiply in the right order.

Matrix powers. If A is square $n \times n$, define $A^0 = I$, $A^1 = A$, and

$$A^k = _____$$

(k factors). Then $A^{k+\ell} = A^k A^\ell$.

Dynamics connection. For $x_{t+1} = Ax_t$ (Lab 4 / Lectures 24–26),

$$x_k = _____.$$

So iterating the recurrence k times is the same as one matrix–vector product with A^k. Powers package "k steps of the system" into a single matrix.

Networks / paths. If A is an adjacency matrix (Boyd: $A_{ij} = 1$ means an edge from j to i), then $(A^k)_{ij}$ counts _____ of length k from j to i. Transition / population models: powers accumulate multi-step flow.

Worked in class

W1. Multistage product. A population moves in two stages each year: migration $B = \begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix}$, then growth/decay $A = \mathrm{diag}(1.05, 0.95)$. Write the one-year matrix M and compute $M \begin{bmatrix} 100 \\ 100 \end{bmatrix}$.

W2. Dynamics as a power. Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0 & 1 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute $x_1 = Ax_0$, $x_2 = Ax_1$, and A^2. Check that $x_2 = A^2 x_0$.

W3. Entry interpretation. Directed graph on nodes $\{1, 2, 3\}$ with edges $1 \to 2$, $2 \to 3$, $3 \to 1$, and also $1 \to 3$. Adjacency (Boyd convention):

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Compute A^2. What does $(A^2)_{31}$ mean? How many length-2 walks from 1 to 3?

W4. Why powers matter for Lab 4. In one sentence: if you already computed A^{10}, how do you get x_{10} from x_0 without looping ten times by hand?

Your turn

Y1. Stage 1: $B = \mathrm{diag}(2, 1)$. Stage 2: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find M for "B then A" and compute $M(1, 1)$.

Y2. With $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (swap coordinates) and $x_0 = (3, 1)$, compute A^2 and $x_2 = A^2 x_0$. Interpret A^2.

Y3. Using the adjacency matrix A from W3, compute $(A^2)_{22}$. Interpret: walks of length 2 that start and end at node 2.

Y4. True or false: for the dynamics $x_{t+1} = Ax_t$, we always have $x_5 = Ax_4$ and also $x_5 = A^5 x_0$.

Answers (Your turn)

Y1. $M = AB = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$. Then $M(1,1) = (-1,2)$.

Y2. $A^2 = I$ (swap twice returns to the start). $x_2 = A^2 x_0 = (3,1) = x_0$. So A^2 is the identity map — two swaps undo each other.

Y3. From W3,

$$A^2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix},$$

so $(A^2)_{22} = 0$: there is no walk of length 2 from 2 back to 2. (From 2 the only edge is $2 \to 3$; from 3 you go to 1, not back to 2 in one more step.)

Y4. **True.** Both follow from $x_{t+1} = Ax_t$ by induction: one step from x_4, and five steps packaged as A^5.